EMORY

Undergraduate Research Programs

Pathways Center

# AI Mathematical Olympiad

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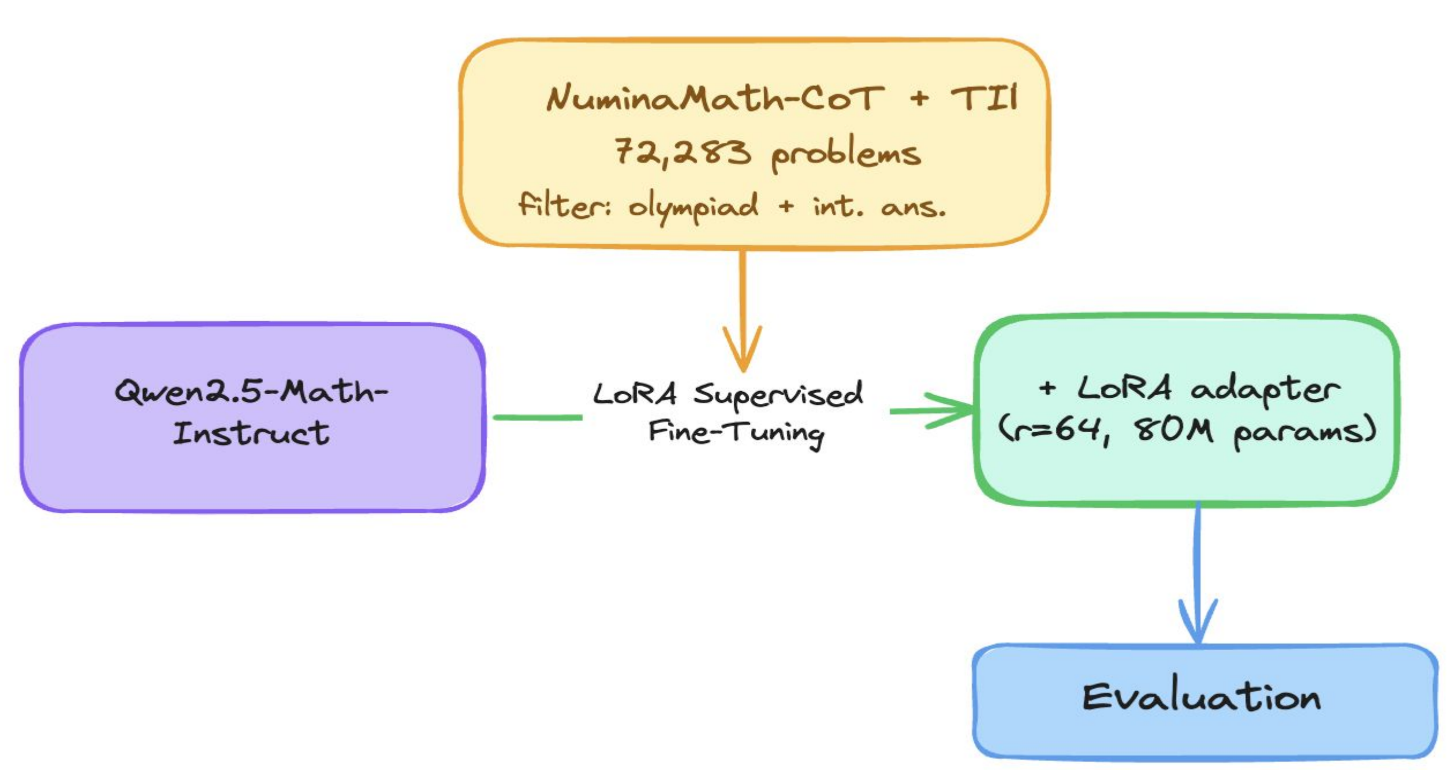
## Introduction

- Large language models now approach human performance on competition math, but state-of-the-art results rely on closed-source systems like GPT-4, Gemini, and Claude.
- The AIMO competition targets open-source models capable of olympiad-level reasoning, where a gap of 30 to 40 accuracy points exists between closed and open systems.
- Closing this gap requires not only better training but also trustworthy evaluation, ensuring measured improvements reflect real capability rather than measurement artifacts.
- Prior work on parameter-efficient fine-tuning such as LoRA has focused primarily on training methodology.
- We argue that when fine-tuning an already-instruction-tuned math model, the evaluation protocol is just as decisive as the training configuration, as a small mismatch in prompt format or output parsing can significantly affect measured results.

## Research Questions

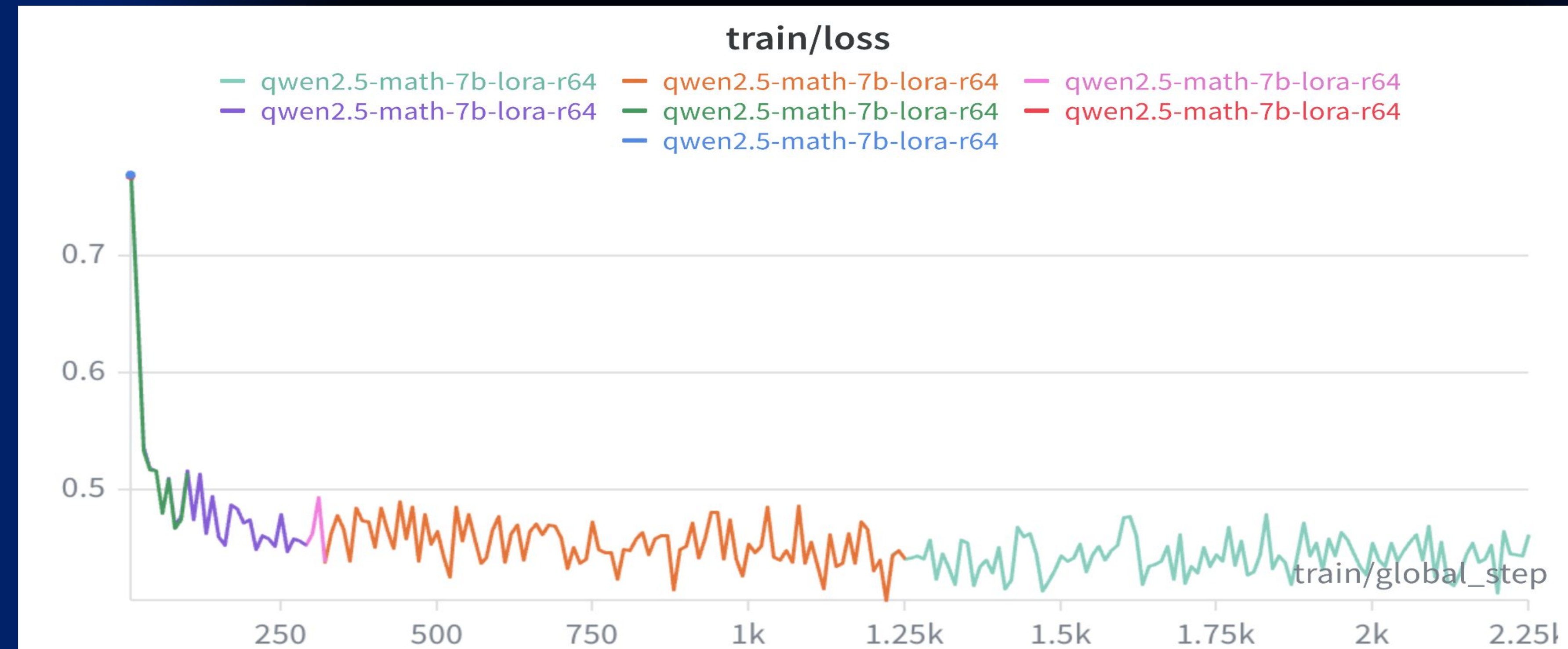
**RQ1.** Does LoRA SFT on olympiad-level CoT data improve an already-instruct-tuned math model, and by how much?
**RQ2.** Under what evaluation conditions does the measured gain appear, and how sensitive is the result to prompt format, generation budget, and answer-matching logic?
**RQ3.** Does the measured in-distribution improvement generalize to held-out olympiad problems (AIME 2024 + 2025)?

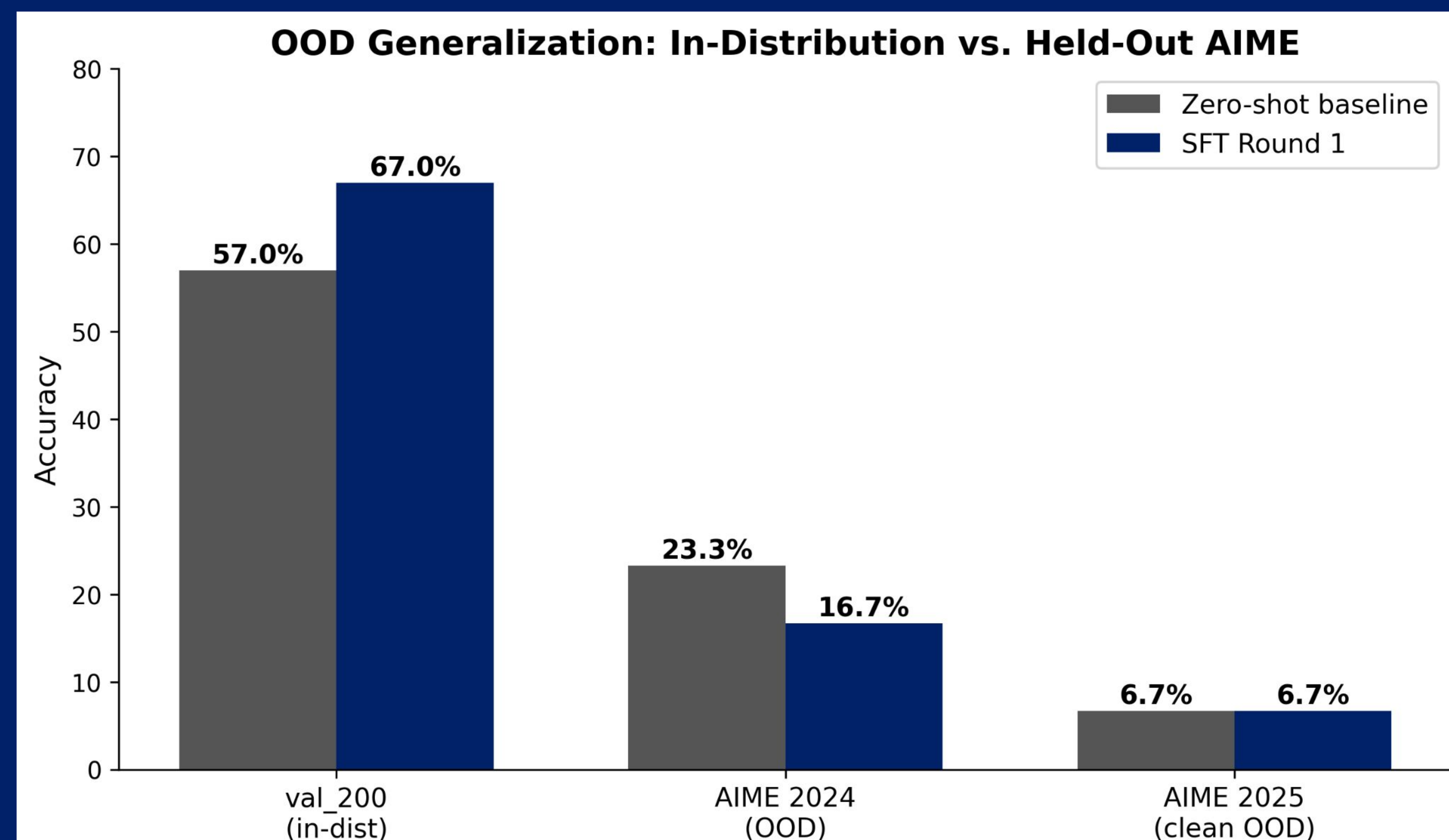
## Methods




We fine-tuned Qwen2.5-Math-7B using LoRA, a parameter-efficient technique that trains a small number of additional weights while keeping the base model frozen. This makes training feasible on a single A100 GPU. Training data was sourced from public math reasoning datasets on HuggingFace and evaluated using exact-match accuracy.

## Results



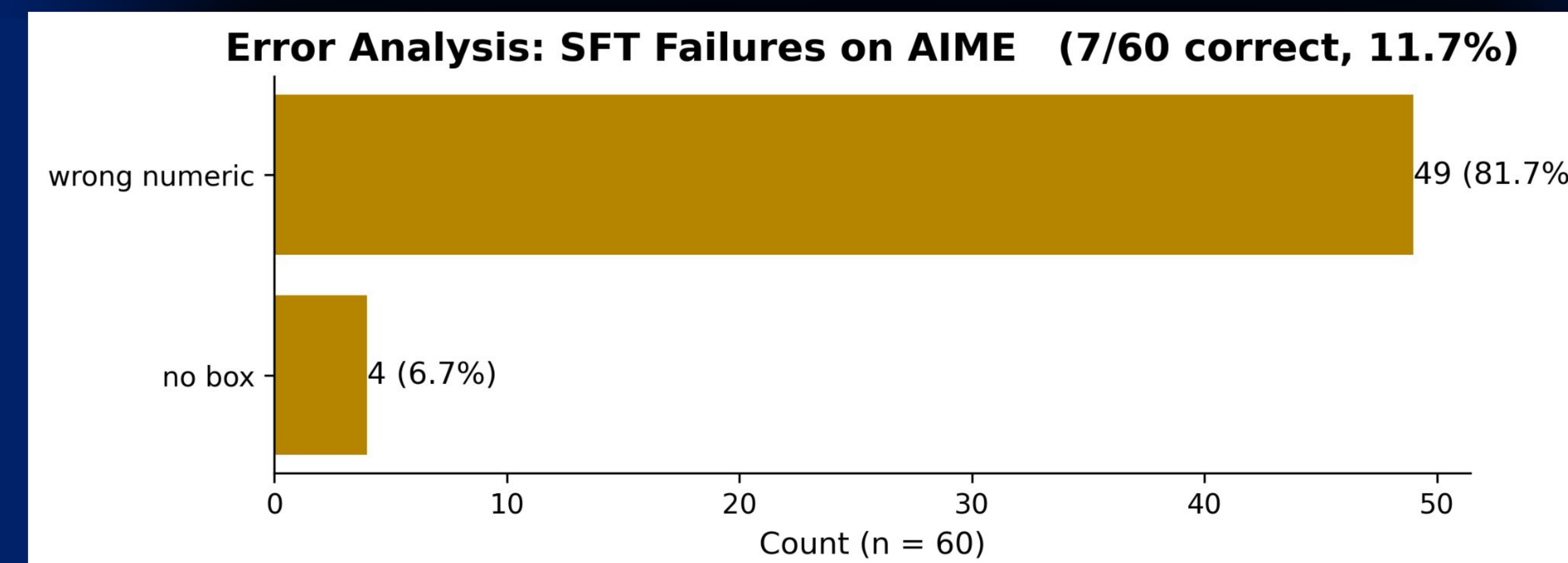

| Benchmark | $n$ | Baseline | SFT | $\Delta$ |
|---|---|---|---|---|
| val_200 (in-distribution) | 200 | 57.0% | **67.0%** | +10.0 |
| AIME 2024 (pre-cutoff) | 30 | 23.3% | 16.7% | −6.7 |
| AIME 2025 (post-cutoff) | 30 | 6.7% | 6.7% | 0.0 |

- Our fine-tuned model improved in-distribution accuracy from 57% to 67%, a 10 point gain over the zero-shot baseline on 200 validation problems.
- The training loss curve confirms successful convergence, dropping sharply from 0.76 in the first 100 steps and stabilizing around 0.45.
- However, gains did not generalize to held-out problems. On AIME 2024 accuracy dropped from 23.3% to 16.7%, and on AIME 2025 both models scored 6.7%.
- These results suggest overfitting to the training distribution, highlighting that out-of-distribution generalization remains a key unsolved challenge.

## Limitations

- **Sample size:** Primary evaluation on 200 problems (±3.4% CI); OOD evaluation on 60 problems (±13% CI). Results on the full 3,805-problem held-out set not yet reported.
- **Training data quality:** NuminaMath-CoT aggregates many sources; we applied structural filters (olympiad + integer answer) but did not verify solution correctness. Some proof-only problems have mislabeled integer "answers."
- **Single training run:** We report one SFT configuration. A full ablation over LoRA rank, LR, and prompt template is future work.
- **Single base model:** Findings may not generalize to non-instruction-tuned or non-math-specialized base models.

## Conclusions/Future Directions




- **What we have achieved:** This project successfully established an end-to-end, memory-efficient Supervised Fine-Tuning pipeline for large language models. By implementing length truncation, dynamic data filtering, and completion-only loss masking, we overcame severe computational bottlenecks and optimized the training of a 7B-parameter model on constrained hardware.
- **What can be improved:** After we test our model on the basic validation set, we also tested our model on advanced validations sets like AIME, where we find our model is performing worse than the baseline. Our analysis to this results has revealed that applying a broader, general-difficulty dataset to an already heavily optimized Instruct model can overwrite its latent, complex reasoning pathways, acting more as a "forgetting" mechanism than an enhancement. To address this, our future improvements will focus on three key areas:
  - **Data-Centric Curation & Strict Difficulty Matching:** Instead of naive data scaling, future iterations will rely on highly purified, smaller subsets (like filtering down to 20k high quality examples). We will ensure the training data strictly aligns with or exceeds the complexity of the target benchmark to prevent diluting the model's capabilities.
  - **Shifting to Base Models:** To stop our models from overthinking or rigidly overfitting to simple formats, future Supervised Fine-Tuning should be conducted on foundational Base models rather than Instruct-tuned models. This preserves the raw deductive potential of the LLM without conflicting with prior alignment training.

## Acknowledgements

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- Compute resources were provided via Google Colab, and training data was sourced from publicly available datasets on HuggingFace.